

AI for Bharat Hackathon

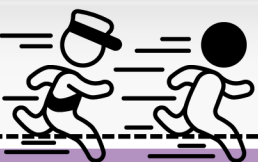
Powered by **aws**



Team Name: AI42J

Team Leader Name: Soumya Gupta

Problem Statement: AI for Communities, Access & Public Impact



PROBLEM

USP&CONTRASTS

FEATURES

USER FLOW

ARCHITECTURE FLOW

TECH STACK

The Problem

26% illiteracy rate turns every government form into an impassable wall

navigating 1000+ state and central initiatives remains a barrier for the most vulnerable.

Shadow lenders use “fine print” as a weapon, secretly inflating debt by 25% to trap the poor in a cycle of modern-day serfdom

90% of new internet users are native speakers; without voice and vernacular support, 900 million people face systemic exclusion.

Our Approach

Proposed Solution

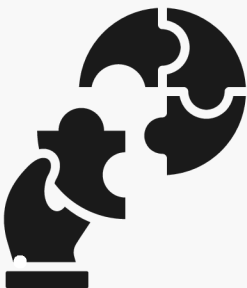
- **VIDHI:** Smart AI platform for government schemes, legal defense, and dispute resolution.
- AI predicts scheme eligibility, analyzes documents, and guides user actions.
- Voice-enabled for all Indians (22+ languages/dialects)
- Voice/efficient data handling reaches low-literacy areas



Introducing **VIDHI**

- VIDHI (Voice-Integrated Defense for Holistic Inclusion) prevents legal exploitation and scheme inaccessibility via AI driven real time guidance.
- Integrates federated learning for privacy-preserving updates, enabling specific warnings for timely intervention and fraud prevention.

Why Solving this Problem is Necessary?



India's untreated legal gaps caused 69,433 bank frauds (2009-2021) and 1.5M aided vs. millions in need



With low scheme awareness nationwide amid over 1,000 initiatives and only 48.8% coverage, showing severe gaps in access.



Our solution protects communities and saves lives with timely intervention.



Using the latest technologies, we created a practical and effective solution.

Problem Addressal

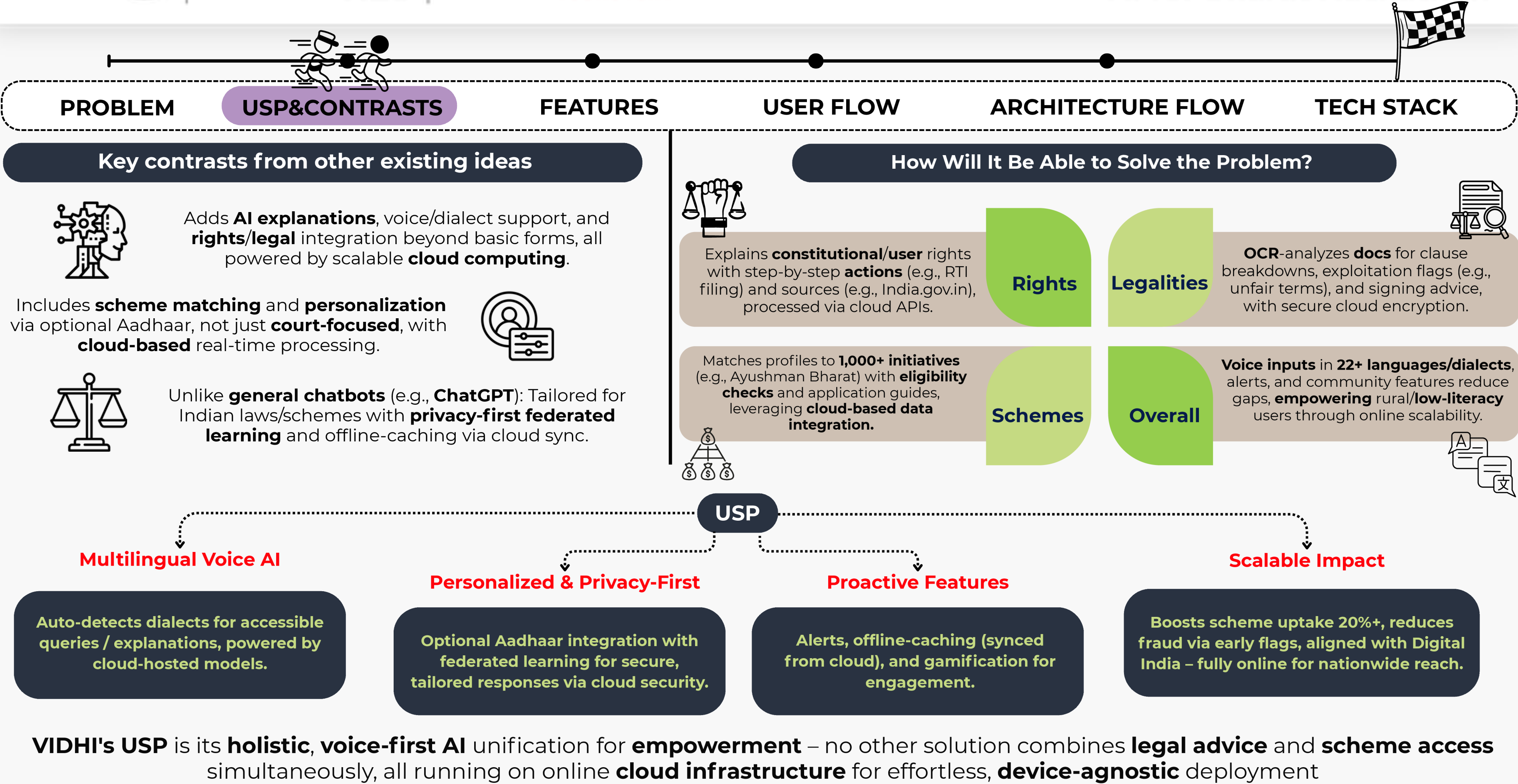


- System Flags unfair clauses early, averting exploitation
- Voice/efficient data handling reaches low-literacy areas
- Dashboards target scheme/rights guidance in hotspots, boosting utilization
- Citizens report issues and get education on everyday rights

Solution Uniqueness



- **Scalable prediction:** Federated learning aggregates anonymized data for personalized rights/scheme searches.
- **Actionable alerts:** Real-time warnings
- **VIDHI** unifies voice-first access in dialects for holistic empowerment.
- **Privacy-First AI :** Shares model updates only—never user data



PROBLEM

USP&CONTRASTS

FEATURES

USER FLOW

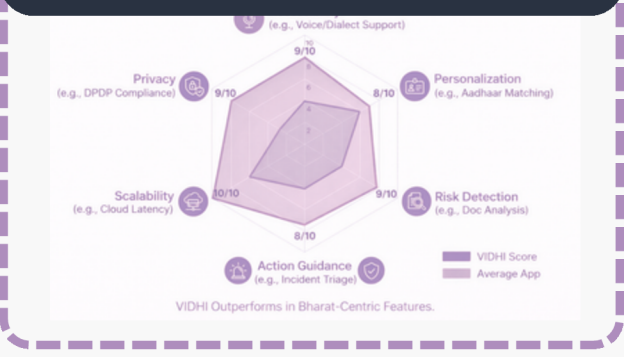
ARCHITECTURE FLOW

TECH STACK

Predictive Scheme Intelligence

Voice queries for scheme details (e.g., **PM Kisan** or **Ayushman Bharat**, retrieved instantly via cloud search)
Step-by-step office/forms guides (location-specific tips)
Auto-match by age/income/gender (personalizes results for accuracy, boosting eligibility success rates)

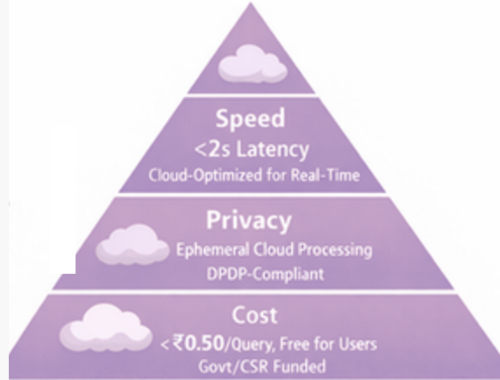
VIDHI Empowerment radar



Real-Time Incident Triage

- **Actions for disputes** (e.g., traffic stops or landlord issues, with legal IPC/CrPC references)
- **Panic mode for arrests** (deploys rights citations)
- **Pre-court resolution steps** (guides low-cost alternatives, reducing escalation to India's 54M+ pending cases)

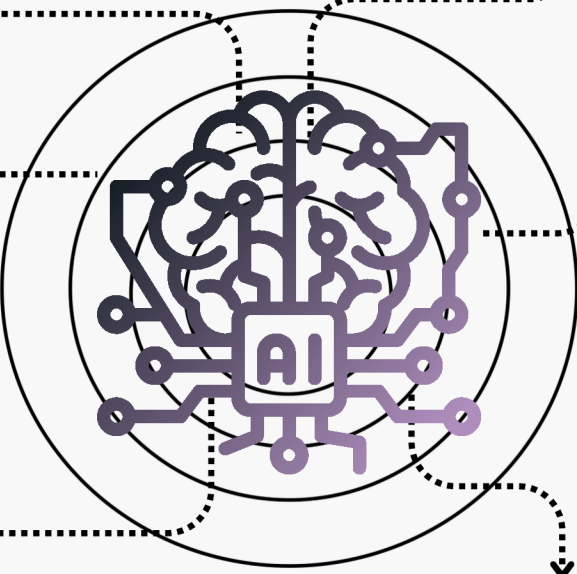
Scale & Trust



VIDHI Input Engine

Voice + OCR inputs (cloud-uploaded for secure processing)
Dialect detection for **personalization**
Aadhaar sync for profile matching (optional, verified via cloud APIs for tailored advice)

Component	Cloud Cost (Annual)
AI Models	₹50K (AWS Bedrock)
Voice APIs	₹30K (Transcribe/Polly)
Data Integration	₹20K (Govt APIs)
Total	₹1L

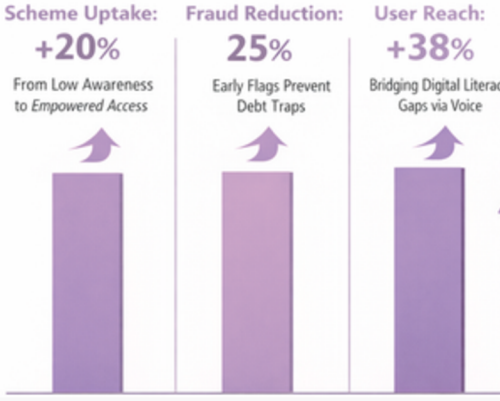


Sahayak Output Layer

- **Voice/text responses** with **alerts** (cloud-generated for quick, actionable insights)
- **Proactive scheme/policy updates** (deadline reminders via push notifications)
- **Community sharing** for **follow-ups** (anonymized forums for learning)

Semantic Contract Analysis

Scans docs for hidden fees (uses **cloud OCR** to detect predatory clauses in loans or rentals)
Flags risks like high interest (provides plain-language warnings e.g., Could add 25% debt - avoid signing)
Advises on **tenancy/land fraud**



Zero-UI Voice Orchestration

Hands-free query to response, **operation** via **cloud speech**, (ideal for low-literacy users)

Dialect audio output (ensuring accessibility across Bharat)
Spoken alerts for **updates** (proactive notifications on schemes/policies, delivered as voice notes)

Cloud Deployment Breakdown:

Component	Annual Cost	Key Benefit
AI Models	₹50K	Scalable Processing
Voice APIs	₹30K	Dialect Support
Data Integration	₹20K	Real-Time Accuracy
Total	₹1L	Nationwide Free Access

PROBLEM

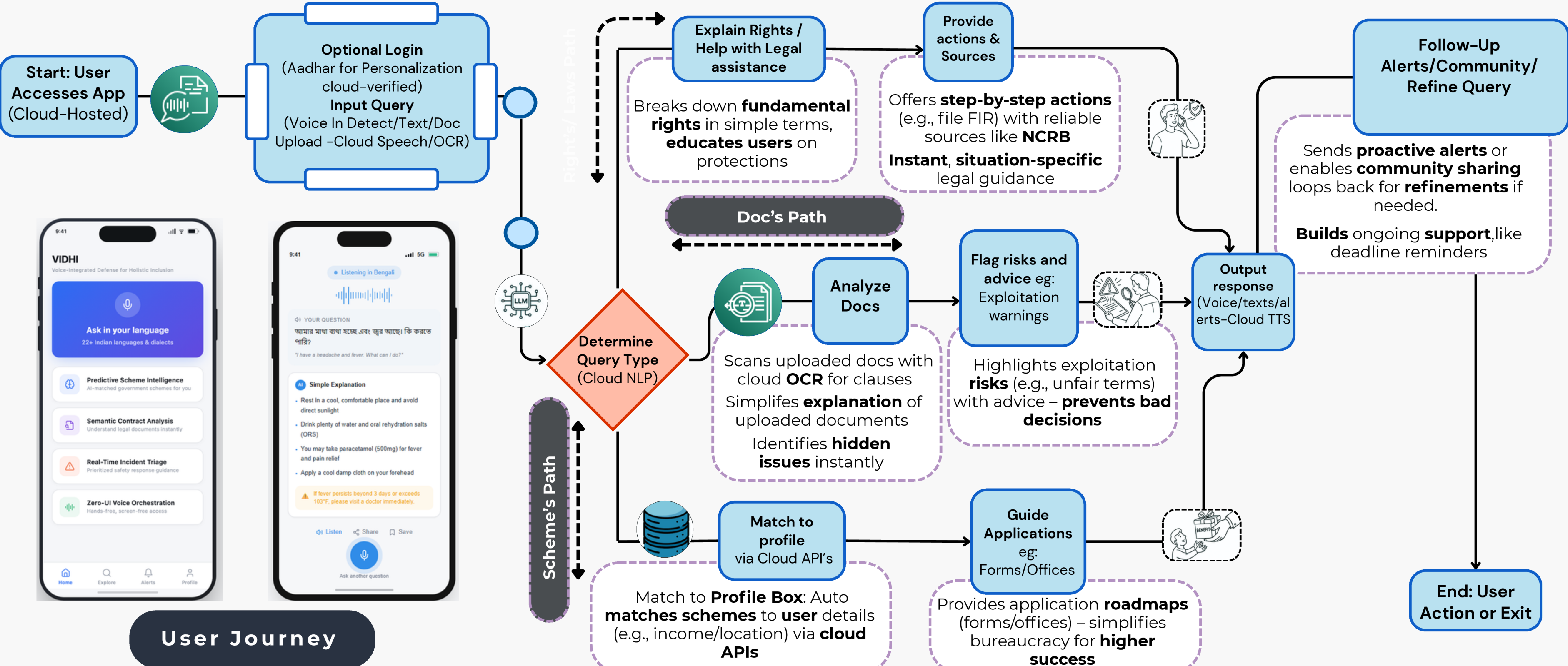
USP&CONTRASTS

FEATURES

USER FLOW

ARCHITECTURE FLOW

TECH STACK



User Journey

PROBLEM

USP&CONTRASTS

FEATURES

USER FLOW

ARCHITECTURE FLOW

TECH STACK

Detects spoken language
routes query accordingly
Enables native response



Language Identification

Identifies user intent
Detects illegal/risky queries
Flags incomplete queries



Intent and Risk Identification

Trained on problem specific data
Handles illegal activity awareness
Provide corrective and safe guidance



Domain Specific AI Engine

Responses in native language
Text-to-speech output
Natural voice output



Language Translation

User



The user gives the
problem input in his
native language

Speech-to-Text



Converts speech to text
Supports Indian language
Optimized for accent

Clarification Loop



Asks follow up questions (if needed)
Requests missing information
Ensures safe understanding

Response Generation



Generates safe explanation
Uses clarified content
Policy-aligned responses

PROBLEM

USP&CONTRASTS

FEATURES

USER FLOW

ARCHITECTURE FLOW

TECH STACK

AI Workflow & Orchestration

- Custom workflow engine
- AI pipelines
- Event-based triggers

AI Intelligence Layer

- OpenAI (LLM)
- BHASHINI APIs
- Prompt orchestration
- AI workflows (multi-step reasoning)

User Interface

- React Native
- TypeScript
- UI localization

Deployment & Scale Infrastructure

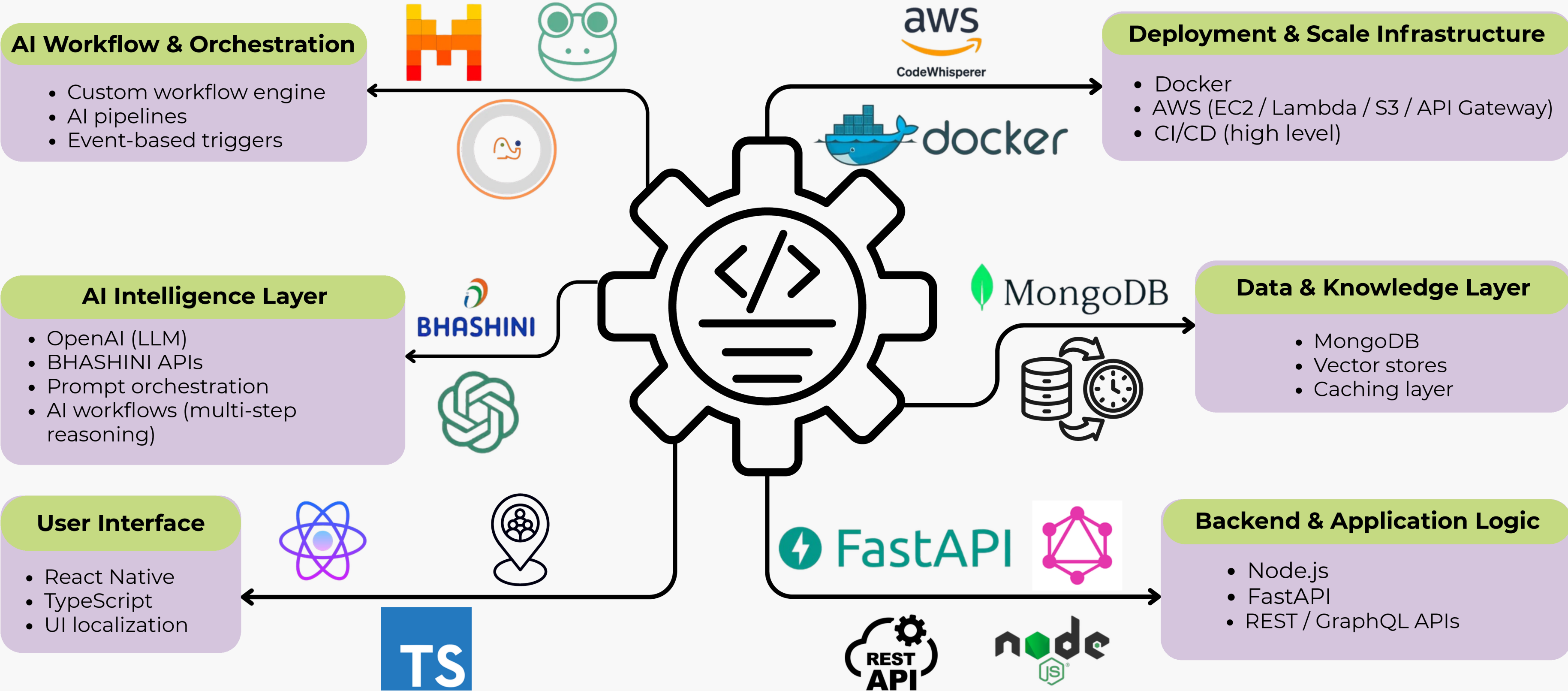
- Docker
- AWS (EC2 / Lambda / S3 / API Gateway)
- CI/CD (high level)

Data & Knowledge Layer

- MongoDB
- Vector stores
- Caching layer

Backend & Application Logic

- Node.js
- FastAPI
- REST / GraphQL APIs



Innovation partner **I12S**

Media partner **YOURSTORY**

AI for Bharat Hackathon

Powered by **aws**

Thank You

